ELECTRONICS WORKSHOP

A PROJECT REPORT ON

LINE FOLLOWING ROBOT

Submitted by

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ABSTRACT

A Line Follower Robot is an autonomous mobile robot designed to follow a predefined path marked by a contrasting line on the ground. The primary objective of this project is to design and implement a compact robotic system capable of detecting and tracking a line using infrared (IR) sensors while maintaining stable movement through controlled motor operation. Such robots demonstrate the fundamental concepts of embedded systems, sensor interfacing, control systems, and autonomous navigation.

The robot uses an array of infrared sensors to detect the difference between the line and the surrounding surface. These sensors continuously monitor the path and send digital signals to the microcontroller based on the reflection of infrared light. In this project, an ESP32 microcontroller is used as the main processing unit due to its high processing speed, multiple GPIO pins, and flexibility for embedded applications. The sensor outputs are processed by the microcontroller to determine the relative position of the line with respect to the robot.

Based on the sensor readings, the microcontroller generates appropriate control signals to drive the motors through an L298N motor driver module. The motor driver controls two DC motors that move the robot forward, left, or right depending on the line position. If the robot deviates from the path, the controller adjusts the speed of the motors using PWM signals to bring the robot back onto the line. This feedback-based control ensures smooth and accurate tracking of the path.

The mechanical structure of the robot consists of a lightweight chassis, two DC geared motors with wheels for movement, a caster wheel for balance, and a battery pack for power supply. The integration of hardware components and embedded programming enables the robot to perform real-time line detection and navigation.

Line follower robots have practical applications in automated transportation systems, industrial material handling, warehouse management, and intelligent vehicle guidance systems. They are widely used in robotics research and education to demonstrate basic automation principles. The successful implementation of this project highlights the importance of sensor-based control and embedded programming in the development of autonomous robotic systems.

INTRODUCTION

Robotics and automation have become an essential part of modern technology, playing a significant role in industries, transportation systems, and everyday applications. Autonomous robots are machines capable of performing tasks without continuous human intervention by sensing their environment and making decisions based on programmed instructions. One of the simplest and most widely studied examples of autonomous mobile robots is the Line Follower Robot.

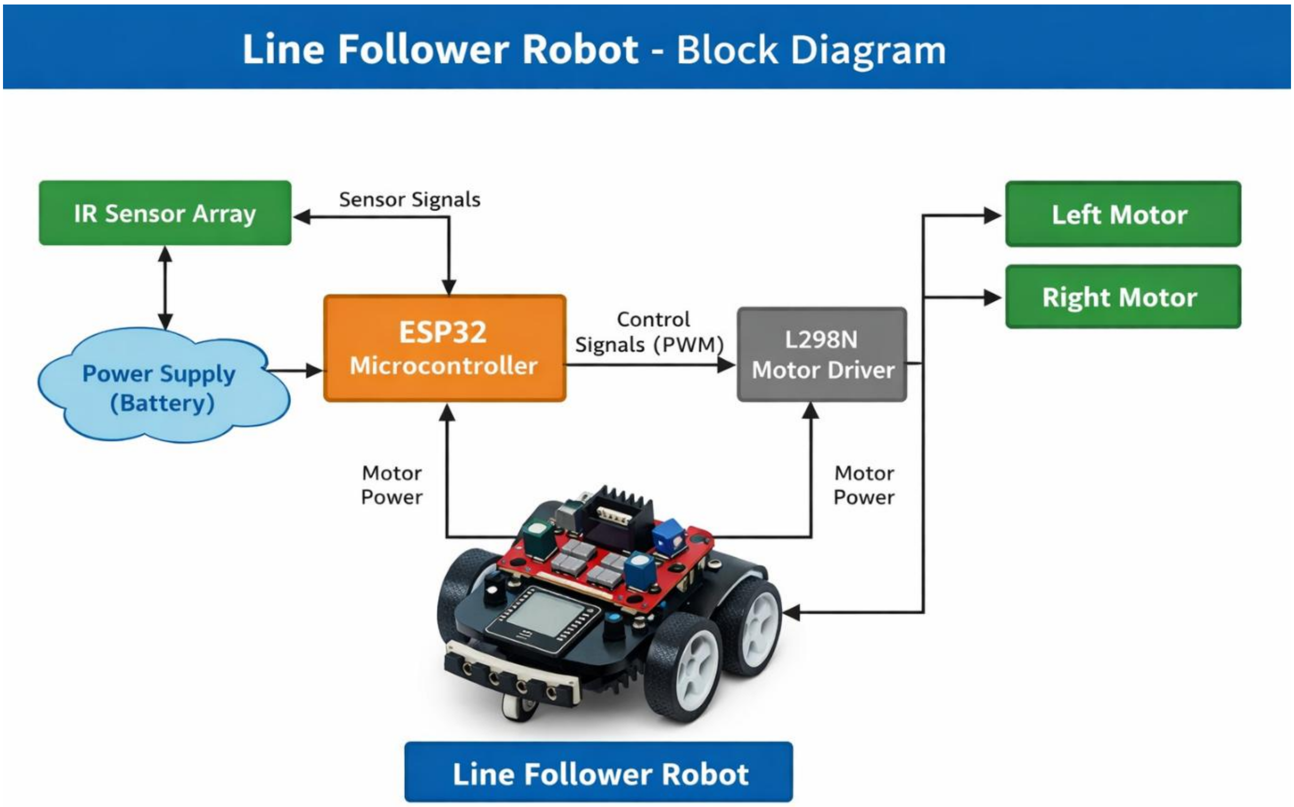
A Line Follower Robot is designed to move along a specific path that is defined by a visible line, usually black on a white surface or white on a black surface. The robot detects this line using infrared (IR) sensors and continuously adjusts its movement to remain on the path. The system operates on the basic principle of light reflection, where the sensor differentiates between the line and the surrounding surface based on the amount of reflected infrared light. This ability allows the robot to determine its relative position with respect to the line.

In this project, the robot is controlled using an ESP32 microcontroller, which acts as the central processing unit of the system. The ESP32 receives input signals from the IR sensor array and processes them according to the programmed logic. Based on the sensor data, the controller generates output signals to drive the motors through an L298N motor driver module. The motor driver regulates the speed and direction of the DC motors, enabling the robot to move forward or adjust its direction to maintain alignment with the path.

The robot typically consists of a chassis, DC geared motors with wheels, a caster wheel for balance, an IR sensor array for line detection, a motor driver circuit, and a rechargeable battery for power supply. Proper integration of these components along with efficient programming enables the robot to navigate the path accurately and respond quickly to changes in the line position.

Line follower robots are widely used as an introductory project in robotics and embedded systems because they demonstrate key concepts such as sensor interfacing, real-time control, and autonomous navigation. Additionally, similar technologies are applied in automated guided vehicles (AGVs), industrial automation, and warehouse transportation systems. Therefore, developing a line follower robot provides valuable practical knowledge in robotics, electronics, and embedded system design.

BLOCK DIAGRAM



1.1 DESCRIPTION OF BLOCK DIAGRAM

The block diagram represents the basic working structure of the line follower robot. The IR sensor array detects the black line on the surface and sends signals to the ESP32 microcontroller. The ESP32 processes these signals and determines whether the robot should move straight, turn left, or turn right.

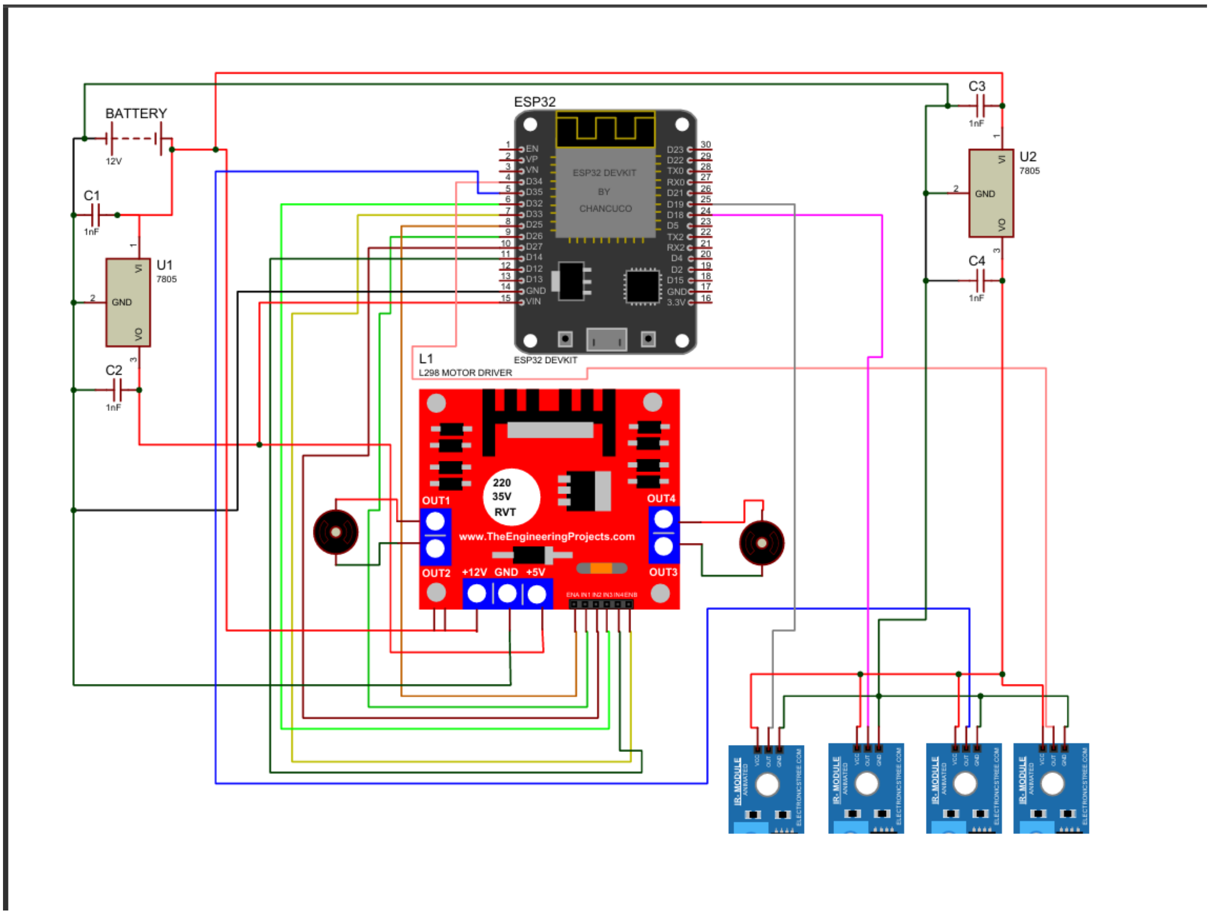
Based on this decision, the ESP32 sends control signals (PWM) to the L298N motor driver, which controls the speed and direction of the left and right DC motors. The motors move the robot accordingly so that it stays on the line.

The power supply (battery) provides electrical power to the sensors, microcontroller, motor driver, and motors for the robot’s operation.

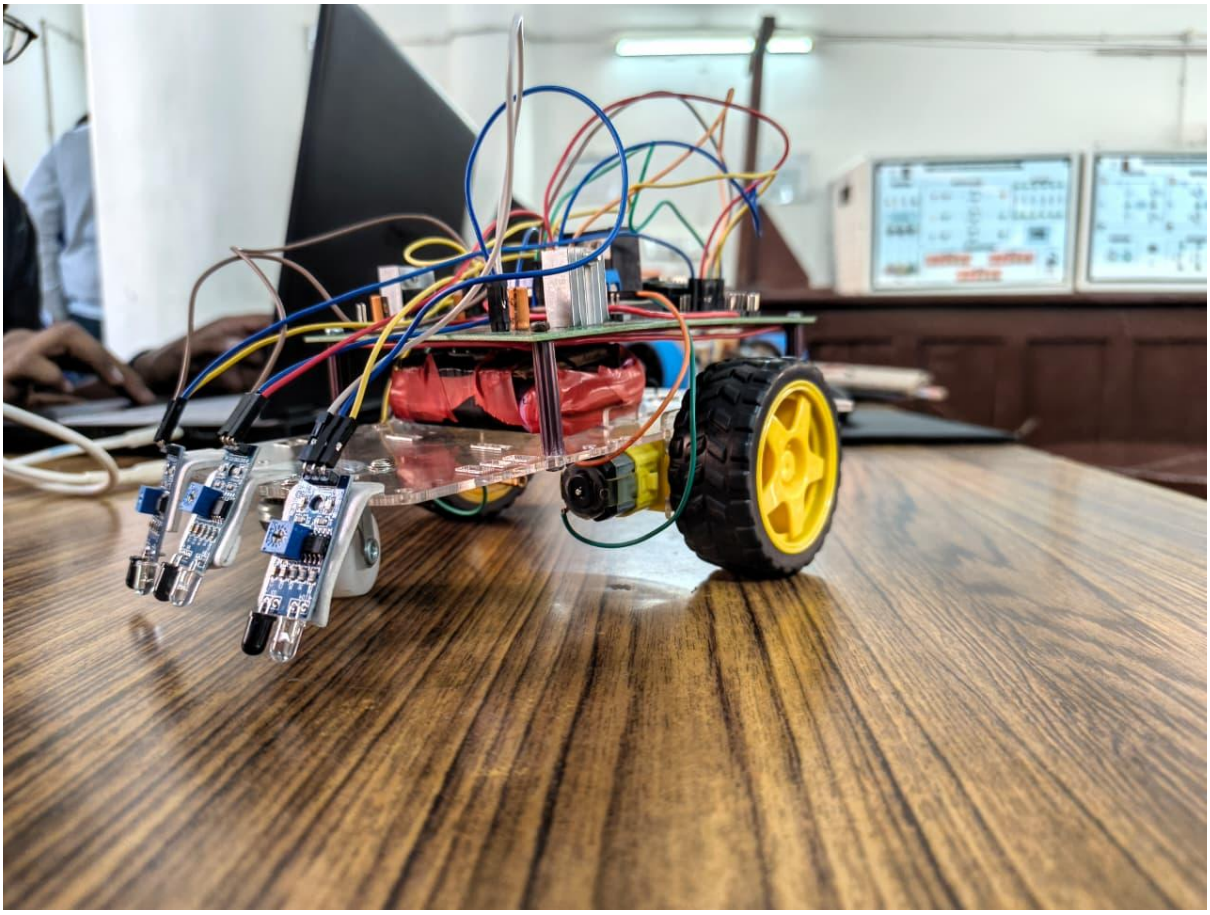
Additionally, the block diagram shows how all components work together as a closed-loop control system. The IR sensors continuously monitor the position of the line and provide real-time feedback to the ESP32 microcontroller. If the robot starts to deviate from the path, the controller immediately adjusts the motor speeds through the motor driver to correct its direction. This continuous sensing, processing, and correction enables the robot to follow the line smoothly and maintain accurate navigation along the predefined path

CHAPTER 2 CIRCUIT DIAGRAM

2.1 CIRCUIT DIAGRAM



CHAPTER 3 IMAGE OF PROJECT

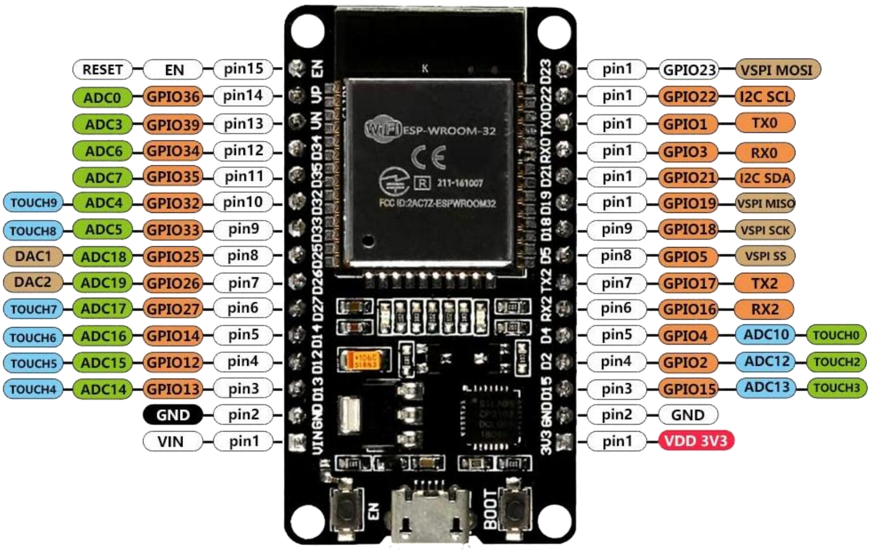


ACTUAL IMAGE OF PROJECT

LIST OF COMPONENTS AND COMPONENTS DESCRIPTION

1) ESP32 Microcontroller Board – Used as the main controller to process sensor data and control the motors.
2) IR Sensor Array TCRT51 – Detects the black line on the surface using infrared reflection.
3) L298N Motor Driver Module – Drives and controls the direction and speed of the DC motors.
4) 7805 IC for Stable 5V output with 12V Input
5) DC Geared Motors (2 Units) – Provide movement to the robot.
6) Robot Wheels (2 Units) – Attached to the motors for motion.
7) Caster Wheel (1 Unit) – Provides balance and support to the robot chassis.
8) Robot Chassis – Base structure to mount all components.
9) Rechargeable Li-ion Battery Pack (3.7V Cells) – Provides power to the system.
10) Battery Holder / BMS Module – Used for safe battery management and power distribution.
11) Connecting Wires (Jumper Wires) – Used for electrical connections between components.
12) Switch – Used to turn the robot ON and OFF.
13) Mounting Hardware (Nuts, Bolts, Spacers) – Used to assemble and secure the components.

ESP 32 Microcontroller



The ESP32 Development Board based on the ESP-WROOM-32 module is a low-cost and powerful microcontroller board designed for IoT and embedded applications. It integrates Wi-Fi and Bluetooth connectivity, a dual-core processor, multiple GPIO pins, and several communication interfaces. The board is widely used for robotics, automation, sensor networks, and smart devices.

- Chip: ESP32 (ESP-WROOM-32 module)
- CPU: Dual-core Xtensa® LX6 microprocessor
- Clock Frequency: Up to 240 MHz
- SRAM: 520 KB
- Flash Memory: Typically, 4 MB external flash
- Wi-Fi:
- IEEE 802.11 b/g/n
- 2.4 GHz frequency band
- Supports Station, SoftAP, and Station+SoftAP modes
- Bluetooth:
- Bluetooth v4.2 BR/EDR
- Bluetooth Low Energy (BLE)
- Operating Voltage: 3.3 V
- Input Voltage (VIN): 5 V via USB or VIN pin
- USB Interface: Micro-USB for programming and power
- Current Consumption:

- Active Wi-Fi: ~80–260 mA
- Deep Sleep: ~10 µA
- Total GPIO Pins: Up to 34 programmable GPIOs
- ADC Channels: 18 channels (12-bit resolution)
- DAC Channels: 2 channels (8-bit)
- Touch Sensors: 10 capacitive touch GPIOs
- PWM Channels: Up to 16 channels

The ESP32 supports multiple communication protocols:

- **UART:** 3 interfaces
- **SPI:** 4 interfaces
- **I2C:** 2 interfaces
- **I2S:** 2 interfaces
- **CAN Bus:** Supported
- **PWM:** LED PWM controller

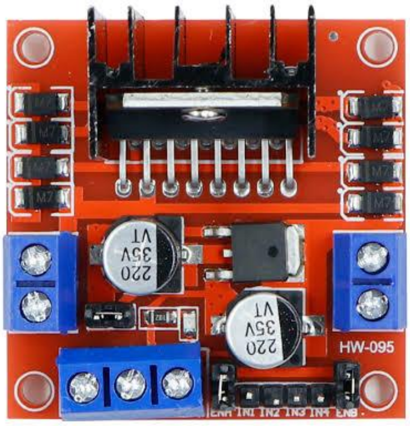
Peripheral Features

- Hall effect sensor
- Temperature sensor (internal)
- Capacitive touch sensing
- RTC with deep sleep support
- Hardware encryption and security features

Development Support

- Programmable using:
- Arduino IDE
- ESP-IDF
- MicroPython
- PlatformIO

L298N Motor Driver Circuit

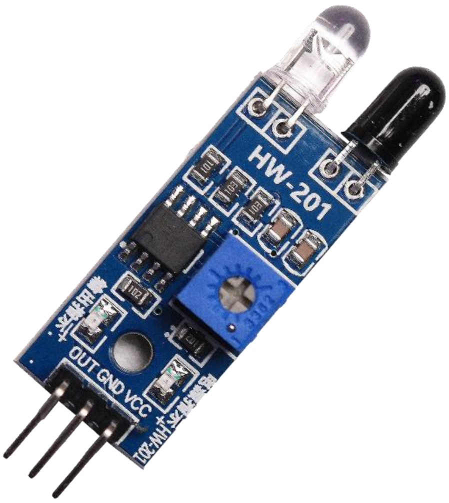


The L298N Motor Driver Module is a high-power motor driver board used to control DC motors and stepper motors. It is based on the L298N dual H-Bridge motor driver IC, which allows control of two DC motors independently or one stepper motor. The module can control the direction and speed of motors using digital signals and PWM from microcontrollers such as ESP32, Arduino, or 8051.

- **Specifications**
- IC Used: L298N Dual H-Bridge Motor Driver
- Driver Type: Dual Full Bridge Driver
- Motor Control: Bidirectional control of two DC motors
- Motor Supply Voltage (Vs): 5V – 35V
- Logic Voltage: 5V
- Maximum Output Current:
- 2A per channel (peak)
- ~1.5A continuous per channel with proper cooling
- Onboard 5V Regulator: Yes (can supply logic circuit when input voltage >7V)
- Number of Channels: 2
- Supported Motors:
- Two DC motors
- One stepper motor
- **Control Pins**
- The module includes the following control pins:

- IN1, IN2: Control direction of Motor A
- IN3, IN4: Control direction of Motor B
- ENA: Enable pin for Motor A (speed control using PWM)
- ENB: Enable pin for Motor B (speed control using PWM)
- **Output Pins**
- OUT1, OUT2: Motor A output
- OUT3, OUT4: Motor B output
- Power Pins
- 12V / VCC: Motor supply input
- GND: Ground
- 5V: Logic power supply
- **Additional Features**
- Onboard heat sink for heat dissipation
- Protection diodes for back EMF from motors
- Screw terminals for easy motor and power connections
- Compatible with PWM speed control
- **Typical Applications**
- Line following robots
- Robot cars and mobile robots
- Motor speed control systems
- Industrial automation projects
- Arduino and ESP32 robotics projects

IR Sensors



The HW-201 IR Sensor Module is an infrared based detection module commonly used for line following robots and obstacle detection systems. It works using an infrared transmitter (IR LED) and an infrared receiver (photodiode/phototransistor). When IR light reflects from a surface or object, the sensor detects the reflection and produces a digital output signal that can be read by a microcontroller such as ESP32, Arduino, or 8051.

The module emits infrared light through the IR LED. When this light hits an object or surface (such as a black or white line), it is reflected back to the IR receiver. The onboard LM393 comparator processes the signal and generates a HIGH or LOW digital output depending on the detected reflection.\

- **Specification**
- Operating Voltage: 3.3V – 5V DC
- Current Consumption: ~20 mA
- Compatible Controllers: ESP32, Arduino, Raspberry Pi, 8051, PIC
- Digital Output (DO)
- Logic signal indicating detection of object or line
- Detection Distance: Approximately 2 cm – 30 cm (adjustable)
- Detection Method: Infrared reflection
- Sensitivity Adjustment: Adjustable using onboard potentiometer
- **Pin Configuration**

The module has three pins:

- VCC: Power supply (3.3V – 5V)
- GND: Ground

- OUT: Digital output signal
- **Onboard Components**
- IR LED (Transmitter)
- Photodiode / IR Receiver
- LM393 Comparator IC
- Sensitivity Adjustment Potentiometer
- Power and Output Indicator LEDs
- **Features**
- Fast response time
- Adjustable sensitivity
- Simple digital output interface
- Compact and lightweight design
- Compatible with most microcontrollers
- **Applications**
- Line follower robots
- Obstacle avoidance robots
- Object detection systems
- Industrial automation sensors
- Robot navigation systems

CODE

```
// Motor Pins
#define ENA 25
#define IN1 26
#define IN2 27

#define ENB 33
#define IN3 32
#define IN4 14

// IR Sensors
#define RIGHT_SENSOR 18
#define MIDDLE_SENSOR 19
#define LEFT_SENSOR 34

int speedA = 90;
int speedB = 90;
void setup()
{
  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(ENB, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  pinMode(RIGHT_SENSOR, INPUT);
  pinMode(MIDDLE_SENSOR, INPUT);
  pinMode(LEFT_SENSOR, INPUT);
}

void loop()
{
  analogWrite(ENA, speedA);
```

```
  analogWrite(ENB, speedB);
  int right = digitalRead(RIGHT_SENSOR);
  int middle = digitalRead(MIDDLE_SENSOR);
  int left = digitalRead(LEFT_SENSOR);
  // Forward
  if(middle == HIGH && left == LOW && right == LOW)
  {
    forward();
  }
  // Sharp Left Turn
  else if(left == HIGH)
  {
    sharpLeft();
  }
  // Sharp Right Turn
  else if(right == HIGH)
  {
    sharpRight();
  }
  else
  {
    forward();
  }
}
void forward()
{
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}
void sharpLeft()
{
```

```
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);   // left motor reverse

  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);    // right motor forward
}
void sharpRight()
{
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);    // left motor forward
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);   // right motor reverse
}
```

CONCLUSION

The Line Follower Robot project demonstrates the practical implementation of autonomous robotic navigation using sensor-based control and embedded systems. The main objective of this project was to design and develop a robot capable of detecting and following a predefined path automatically without human intervention. By integrating sensors, a microcontroller, and motor control circuitry, the robot was able to successfully detect the line and adjust its movement accordingly to remain on the path.

In this project, the IR sensor array played a crucial role in detecting the difference between the line and the surrounding surface based on infrared light reflection. The sensor outputs were continuously monitored by the ESP32 microcontroller, which processed the signals and determined the position of the line relative to the robot. Based on this information, the controller generated appropriate control signals to the L298N motor driver module. The motor driver then controlled the direction and speed of the DC motors, enabling the robot to move forward, turn left, or turn right depending on the path condition. This real-time sensing and control mechanism allowed the robot to follow the line smoothly and accurately.

The project also helped in understanding the importance of proper hardware integration and programming in embedded systems. The successful functioning of the robot depends on correct sensor positioning, efficient programming logic, stable motor control, and proper power management. By carefully assembling and interfacing these components, the robot was able to perform its task reliably. This project provided valuable hands-on experience in areas such as sensor interfacing, motor driver control, microcontroller programming, and robotic system design.

Moreover, the concept of a line follower robot has several real-world applications in industries and automation systems. Similar technologies are used in automated guided vehicles (AGVs) for material transportation in factories and warehouses. They are also used in robotic research, automated delivery systems, and intelligent transportation systems. These applications highlight the significance of autonomous robots in improving efficiency, reducing human effort, and increasing operational accuracy.

In conclusion, the Line Follower Robot project successfully demonstrates the principles of robotics, automation, and embedded system design. The project not only enhances theoretical understanding but also develops practical skills in electronics and programming. It serves as a strong foundation for more advanced robotic systems and encourages further exploration in the field of intelligent automation and robotic technology.